**Week 3**

# Agile frameworks

## Agile framework:

There are some agile methodologies and frameworks such as Scrum, XP, Battlefield Agility, Safe, Nexus, Kanban, DSDM, DAD etc. which can be practiced to effectively follow the principles and values of agile methodologies.

## Agile ceremonies:

- **Sprint Planning**

  Sprint Planning is used to determine what the team will accomplish in the upcoming Sprint. The event itself has two parts. The first half of Sprint Planning is used to determine 'What' the team will be working on, by pulling items from their Backlog into their Sprint Backlog. The second half of Sprint Planning is when the Development Team determines 'How' they will accomplish the work that's been pulled into the Sprint Backlog

- **Daily Scrum**

  The Daily Scrum, sometimes referred to as the Daily Standup, has a time-box for 15 minutes or less, and is specifically for the benefit of the development team. The goal of this event is for the team to get in sync on a daily basis, allowing for better collaboration and transparency. The Daily Scrum should be held at the same time each day and should not include anyone outside of the Scrum Team.

- **Sprint Review**

  The Sprint Review is when the team presents their work from the Sprint to the project's stakeholders. It should cover not only the work they accomplished, but also open discussions around the work they were not able to complete.

- **Sprint Retrospective**

  The Sprint Retrospective is the primary event in which the Scrum Team can inspect and adapt their approaches based on their experiences from the previous sprints.

## Roles and Responsibilities of Agile:

An Agile team working in Scrum has three roles:

- **Product Owner** – Often an executive or key stakeholder, the Product Owner has a vision for the end product and a sense of how it will fit into the company's long-term goals. This person will need to direct communication efforts, alerting the team to major developments and stepping in to course-correct and implement high-level changes as necessary.

- **Scrum Master** – The Scrum Master is most akin to a project manager. They are guardians of process, givers of feedback, and mentors to junior team members. They oversee day-to-day functions, maintain the Scrum board, check in with team members, and make sure tasks are being completed on target.
- **Team Member** – Team members are the makers: front- and back-end engineers, copywriters, designers, videographers, you name it. Team members have varied roles and skills but all are responsible for getting stuff done on time and in excellent quality.

**Extreme Programming:**

Extreme Programming or XP which it is informally called, is the most well-known of the agile methods. XP consists of a set of engineering practices for a team to apply when developing software. These practices are based on commonly accepted software development approaches, but what makes XP extreme in some sense is that these approaches are applied at an extreme level.

**Extreme programming (XP)** is one of the most important software development frame work of Agile models. It is used to improve software quality and responsive to customer requirements. The extreme programming model recommends taking the best practices that have worked well in the past in program development projects to extreme levels.

**Good practices needs to practiced extreme programming(XP practices):** Some of the good practices that have been recognized in the extreme programming model and suggested to maximize their use are given below:

- **Code Review:** Code review detects and corrects errors efficiently. It suggests pair programming as coding and reviewing of written code carried out by a pair of programmers who switch their works between them every hour.
- **Testing:** Testing code helps to remove errors and improves its reliability. XP suggests test-driven development (TDD) to continually write and execute test cases. In the TDD approach test cases are written even before any code is written.

- **Incremental development:** Incremental development is very good because customer feedback is gained and based on this development team come up with new increments every few days after each iteration.
- **Simplicity:** Simplicity makes it easier to develop good quality code as well as to test and debug it.
- **Design:** Good quality design is important to develop a good quality software. So, everybody should design daily.
- **Integration testing:** It helps to identify bugs at the interfaces of different functionalities. Extreme programming suggests that the developers should achieve continuous integration by building and performing integration testing several times a day.

**Basic principles of Extreme programming:** XP is based on the frequent iteration through which the developers implement User Stories. User stories are simple and informal statements of the customer about the functionalities needed.

**Some of the basic activities that are followed during software development by using XP model are given below:**

- **Coding:** The concept of coding which is used in XP model is slightly different from traditional coding. Here, coding activity includes drawing diagrams (modeling) that will be transformed into code, scripting a web-based system and choosing among several alternative solutions.
- **Testing:** XP model gives high importance on testing and considers it be the primary factor to develop a fault-free software.
- **Listening:** The developers needs to carefully listen to the customers if they have to develop a good quality software. Sometimes programmers may not have the depth knowledge of the system to be developed. So, it is desirable for the programmers to understand properly the functionality of the system and they have to listen to the customers.
- **Designing:** Without a proper design, a system implementation becomes too complex and very difficult to understand the solution, thus it makes maintenance expensive. A good design results elimination of complex dependencies within a system. So, effective use of suitable design is emphasized.
- **Feedback:** One of the most important aspects of the XP model is to gain feedback to understand the exact customer needs. Frequent contact with the customer makes the development effective.
- **Simplicity:** The main principle of the XP model is to develop a simple system that will work efficiently in present time, rather than trying to build something that would take time and it may never be used. It focuses on some

specific features that are immediately needed, rather than engaging time and effort on speculations of future requirements.

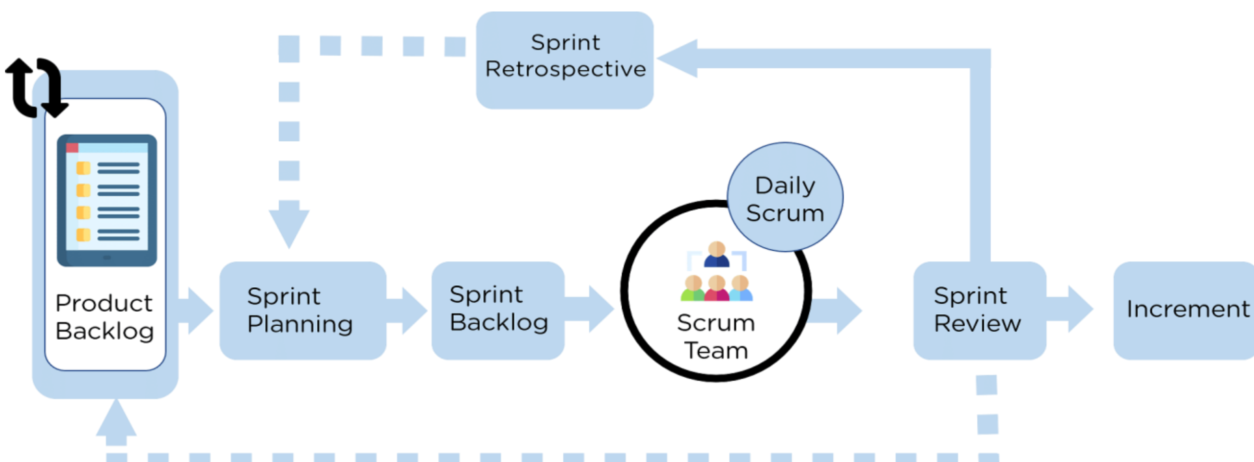
**Applications of Extreme Programming (XP):** Some of the projects that are suitable to develop using XP model are given below:

- **Small projects:** XP model is very useful in small projects consisting of small teams as face to face meeting is easier to achieve.
- **Projects involving new technology or Research projects:** This type of projects face changing of requirements rapidly and technical problems. So XP model is used to complete this type of projects.

## Scrum:

Scrum is a popular framework that enables teams to work together. Based on Agile principles, Scrum enables the development, delivery, and sustenance of complex projects. It enables teams to hypothesize how they think something works, try it out, learn and reflect from their experiences, and make appropriate changes. Scrum is a set of practices and roles which helps manage a software development project.

## Scrum Framework:




### ❖ Product Backlog

The first step is to create a list of tasks that need to be completed to achieve the requirements of the stakeholders/ clients.

### ❖ Sprint Planning

During this stage, the team determines the tasks from the product backlog that they want to work towards completing during the sprint

- **Sprint Backlog**

  The tasks discussed during the sprint planning are added to the sprint backlog

- **Scrum Team**

  The Scrum team (usually consists of 5 to 9 members) works on the tasks mentioned in the sprint backlog.

- **Daily Scrum**

  The team will have daily Scrum meetings, which are 15-minute sessions, during which the team members synchronize their activities and plan their activities for the day.

- **Sprint Review**

  After a sprint is completed, a sprint review takes place. Involving the team, scrum master, product owner, and stakeholders, the sprint review shows what the team accomplished during the sprint. During the meeting, questions are asked, observations are made, feedback and suggestions are also given
.

- **Product Backlog**

  At this point, the product owner presents the product backlogs to the stakeholders for suggestions for tasks that can be added in the upcoming sprints, and so on.

- **Sprint Retrospective**

  After the sprint review, the sprint retrospective takes place. During this meeting, past mistakes, potential issues, and new ways to handle them are identified. Data from here is incorporated when planning the new sprint.

- **Increment**

  A workable output is provided to the stakeholders.

**<u>Scrum Ceremonies:</u>**

- **Sprint Planning Meeting**

  Sprint is a fixed length iterations cycle in which the development works happens, and at the end of the sprint the team delivers working software which can be shipped to the customer. So in the beginning of every Sprint cycle an important meeting held to define the goals of the sprint, which is called sprint planning meeting.

- **Daily Stand-up Meeting**

  The daily scrum meeting is often called the daily stand-up, in which the entire scrum teams comes together for no more than 15 minutes.
- **Sprint Review Meeting**

  The sprint review is one of the most important ceremonies in Scrum where the team gathers to review completed work and determine whether additional changes are needed.
- **Sprint Retrospective Meeting**

  The sprint retrospective is a recurring meeting held at the end of a sprint used to discuss what went well during the previous sprint cycle and what can be improved for the next sprint. The Agile sprint retrospective is an essential part of the Scrum framework for developing, delivering, and managing complex projects.

**<u>Scrum Artifacts:</u>**

Scrum artifacts are the main components of the Scrum process. These artifacts enable you to improve transparency and the team's understanding of the work that they're doing. The artifacts are:

1. **Product Backlog**

   The product backlog consists of a list of new features, the changes made to existing features, bug fixes, changes to the infrastructure, and other artifacts that need to be completed to ensure the team satisfies a particular requirement. It is a source of all things the team works on

.

2. **Sprint Backlog**

   The sprint backlog is a subset of the product backlog that contains tasks that the team aims to complete to satisfy goals that were decided based on negotiations between the product owner and the team. Tasks are identified from the product backlog and are added to the sprint backlog.

3. **Product Increment**

   The product increment is a collection of all the product backlog items tasks completed as part of a sprint and the value of the increments of earlier sprints. Increments refer to inspectable and usable work done at the end of the sprint. It represents a step towards the overall goal of the organization. The outcome must be in functional condition, even if the product owner doesn't decide to release it.